

SIMATS ENGINEERING

ASSESSMENT TOOL 1

Cryptography and Network Security

CO1 – Application-Based Questions (Q2 and Q4)

Course Code	CSA51
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Questions Attempted

Question 2 – Intercepted Suspicious Message. A cybersecurity intern observes the ciphertext “GSRH RH Z HVXIVG” during network monitoring. Its structure indicates that a basic substitution method may have been used. Determine the likely plaintext and explain the reasoning using word patterns, repeated letters, and characteristics of classical ciphers.

Question 4 – Suspicious Digital Asset Investigation. During a forensic examination, a company logo appears ordinary, but investigators suspect that confidential information may be concealed inside the image. Explain how the presence of hidden information can be identified and describe two methods that can make such hidden communication more resistant to detection or extraction.

SIMATS ENGINEERING
ASSESSMENT TOOL 1
Cryptography and Network Security

Code of Conduct

I, **Dudekula Mohammad Ilyas** (Name / Reg No), certify that this submission represents my own work and that I have followed the academic guidelines prescribed for this assessment. I understand that failure to comply with academic integrity requirements may lead to disciplinary action.

Signature of the student: Ilyas

Student Details

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Register Number	192325017
Course	CSA51 - Cryptography and Network Security
Assessment	Assessment Tool 1 - CO1 (Application-Based Questions)

Question 2: Intercepted Suspicious Message

Ciphertext: GSRH RH Z HVXIVG (word lengths: 4, 2, 1 and 6; 13 letters in total).

The preserved spaces and simple letter replacement suggest a monoalphabetic substitution cipher. Two common possibilities are a Caesar shift and the Atbash transformation.

SIMATS ENGINEERING

ASSESSMENT TOOL 1

Cryptography and Network Security

(a) Most Likely Plaintext

The ciphertext is most consistently explained by the Atbash cipher. In this method, the alphabet is reflected around its midpoint: A maps to Z, B maps to Y, C maps to X, and so forth. Using A = 0 through Z = 25, decryption can be written as:

$$P = (25 - C) \bmod 26$$

Because the same transformation works in both directions, Atbash is self-inverse and does not require a separate key.

A	B	C	D	E	F	G	H	I	J	K	L	M	
Z	Y	X	W	V	U	T	S	R	Q	P	O	N	
N	O	P	Q	R	S	T	U	V	W	X	Y	Z	

Atbash mapping (selected): A↔Z, B↔Y, C↔X, D↔W, E↔V, F↔U, G↔T, H↔S, I↔R, J↔Q, K↔P, L↔O, M↔N.

Letter-by-letter Decryption

Cipher	C	25 - C	Plain
G	6	19	T
S	18	7	H
R	17	8	I
H	7	18	S
R	17	8	I
H	7	18	S
Z	25	0	A
H	7	18	S
V	21	4	E
X	23	2	C
I	8	17	R
V	21	4	E
G	6	19	T

Therefore: GSRH = THIS, RH = IS, Z = A, and HVXIVG = SECRET.

Answer: The recovered message is “THIS IS A SECRET”, obtained through the Atbash substitution.

SIMATS ENGINEERING

ASSESSMENT TOOL 1

Cryptography and Network Security

(b) Reasoning and Verification

1. Word-length and pattern check

A substitution cipher normally preserves spaces and therefore keeps the original word lengths. Here the pattern is 4–2–1–6. Since English has only A and I as common one-letter words, the single cipher character Z is likely to represent one of them. The repeated two-letter sequence RH also appears at the end of the first word, which strongly supports the pattern THIS IS A SECRET.

2. Repeated-character evidence

The pair RH occurs both as the final two letters of GSRH and as the complete second word. Under a monoalphabetic mapping, a repeated plaintext sequence must remain repeated in the ciphertext. Likewise, V occurs twice in HVXIVG, corresponding to the two E characters in SECRET.

3. Frequency comparison

The ciphertext character counts are consistent with the proposed plaintext. H occurs three times, while G, R and V occur twice each; the remaining relevant symbols occur once. After decryption, these map to S, T, I and E respectively. The frequency structure therefore remains consistent with a monoalphabetic substitution.

4. Checking the Caesar-cipher possibility

A Caesar cipher uses one fixed alphabetic shift. Testing the shifts implied by the single-letter word does not produce meaningful English, whereas reversing the alphabet gives readable words throughout. This makes Atbash the consistent explanation for the complete ciphertext.

5. Re-encryption check

Applying $C = 25 - P$ to the proposed plaintext reproduces the original ciphertext. For example, T maps to G, H to S, I to R and S to H, giving GSRH. Similarly, IS becomes RH, A becomes Z, and SECRET becomes HVXIVG. The exact match verifies the decryption.

6. Atbash characteristics

Atbash is a fixed monoalphabetic substitution, has no independent secret key, is self-inverse, and preserves word boundaries and the frequency profile. These properties also explain why pattern and frequency analysis can reveal the plaintext relatively easily.

Conclusion: The word structure, repeated sequences, frequency relationship, Caesar-shift comparison and successful re-encryption together support the conclusion that the ciphertext uses Atbash and decodes to “THIS IS A SECRET”.

SIMATS ENGINEERING

ASSESSMENT TOOL 1

Cryptography and Network Security

Question 4: Suspicious Digital Asset Investigation

A normal-looking company logo can potentially carry concealed information through steganography. In this approach, a cover image is modified so that a secret payload is embedded while the visible appearance remains essentially unchanged. This differs from cryptography: encryption hides the meaning of a message, whereas steganography attempts to hide the existence of the message itself.

How Image Data Can Be Concealed

A widely used method is Least Significant Bit (LSB) embedding. The least significant bit of colour values is changed to carry payload bits. Since a pixel value changes by at most one unit in the 0–255 range, these modifications are generally difficult to notice visually. Other possibilities include appending information after a file's normal end marker, placing data in metadata, or modifying JPEG DCT coefficients.

Byte	Original	Secret bit	Modified
1	11001010	0	11001010
2	10110100	1	10110101
3	01101101	0	01101100
4	10011000	0	10011000
5	11100110	1	11100111
6	01010011	0	01010010
7	10101101	0	10101100
8	00111010	0	00111010

Example: The table illustrates how the eight bits of the letter H (01001000) can be distributed across eight carrier bytes.

(a) Detecting Concealed Data

1. Preserve & hash	2. Inspect file	3. Statistical tests	4. Tool-assisted analysis	5. Validate & report
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SIMATS ENGINEERING

ASSESSMENT TOOL 1

Cryptography and Network Security

Step-by-step forensic detection

Step 1 – Secure acquisition

Make a forensic copy of the logo, preferably using a write-blocking process, calculate a SHA-256 hash, and maintain the chain of custody. The analysis should be performed on the working copy. Comparison with the organisation's genuine logo can reveal whether the file has been altered.

Step 2 – File structure and metadata examination

Compare the file size with an authentic image of the same dimensions. Inspect the bytes following the normal JPEG end marker (FF D9) or PNG IEND chunk, because appended content can be invisible to image viewers. File-signature carving can identify embedded ZIP, PDF or text data. Metadata and readable strings can also expose unusual comments, software tags or hidden text.

Step 3 – Visual and statistical examination

Inspect individual bit planes, particularly the LSB plane, for unexpected structure or noise. Histogram analysis can reveal unusual relationships between neighbouring pixel values. Chi-square, RS and sample-pair analysis can estimate whether LSBs have been systematically altered. For JPEG files, abnormal DCT coefficient patterns may also indicate embedding.

Step 4 – Tool-supported analysis

Common forensic utilities can assist different parts of the examination. ExifTool, strings and xxd can inspect metadata, text and raw bytes; Binwalk and Foremost can carve embedded files; Stegsolve and zsteg can inspect bit planes and common LSB patterns; StegExpose and Aletheia can perform statistical steganalysis; Steghide or Stegseek can be used when authorised extraction is required.

Step 5 – Validate and document

Recovered content should be hashed and recorded together with the tools and observations that produced it. If statistical evidence suggests steganography but no payload can be recovered, the finding should be reported as probable steganography rather than as a confirmed extraction.

(b) Two Methods for Strengthening Hidden Communication

Technique 1 – Encrypt and compress before embedding. The secret content can first be compressed and then encrypted with a strong authenticated encryption method such as AES-256-GCM. A password-derived key can be generated using a suitable KDF such as PBKDF2 or Argon2. The encrypted output is then embedded. If the hidden data is extracted, it remains unreadable without the key, and the ciphertext does not expose ordinary file signatures or meaningful text.

Technique 2 – Key-controlled adaptive low-rate embedding. Instead of placing payload bits sequentially, a secret stego-key can control pseudo-random pixel selection. Embedding can favour textured or naturally noisy regions instead of smooth areas. Keeping the payload small and using efficient matrix/error-correcting techniques can reduce the number of altered pixels and therefore reduce statistical distortion.

SIMATS ENGINEERING

ASSESSMENT TOOL 1

Cryptography and Network Security

Effect of the Two Protection Techniques

Resistance to detection. Scattering changes across textured regions avoids obvious sequential LSB patterns and reduces the likelihood of strong histogram or flat-region anomalies. A low embedding rate further limits measurable distortion.

Resistance to extraction. Without the secret stego-key, an investigator does not know which pixels carry the payload or the order in which they should be interpreted. Consequently, a basic LSB reader is unlikely to recover the intended data.

Overall conclusion. A suspected stego image should be examined through evidence preservation and hashing, file-structure and metadata checks, bit-plane inspection, statistical analysis and specialised forensic tools. For protecting hidden communication, encrypting the payload before embedding and using key-controlled, adaptive, low-rate embedding provide complementary protection: the data is difficult to recognise statistically and remains unreadable if it is recovered.

Assessment Rubric

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25	Shows strong conceptual understanding and integrates multiple relevant concepts/technologies.	Demonstrates good understanding and integrates most concepts appropriately.	Shows basic understanding with limited integration.	Shows weak understanding and cannot effectively connect the concepts.
Experimental Design	30	Uses a systematic, well-planned approach with an efficient solution.	Implementation is correct with only minor issues.	Basic implementation is present but includes errors or inefficiencies.	Implementation is incorrect or incomplete.
Use of Tools	20	Uses tools and technologies effectively with suitable configuration.	Uses appropriate tools and executes them correctly.	Tool usage is limited or inconsistent.	Uses tools incorrectly or minimally.
Analysis of Results	15	Provides accurate interpretation, validation and meaningful analysis.	Provides reasonable interpretation and analysis.	Analysis is basic and provides limited insight.	Analysis is incorrect or insufficient.
Documentation	10	Produces clear, organised documentation supported by diagrams, code and results.	Documentation is organised and includes necessary information.	Documentation is basic and misses some details.	Documentation is incomplete or poorly organised.